

Dr Waseem Raza, PhD

Machine Learning Engineer | AI, Wireless Systems & Research

Applied AI | Generative AI | Wireless Networks | Research and Engineering

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Professional Summary

AI/ML and Generative AI Engineer with a PhD in Electrical and Computer Engineering and 7+ years of experience developing data-driven systems across enterprise, industry, and research environments. Experience includes LLM-enabled analytics, AI-ready data pipelines, domain-adaptive machine learning, and scalable cloud solutions using Python, SQL, Azure Databricks, Apache Spark, and Azure services. At AT&T, lead development and operational integration of AI solutions for critical service-assurance workflows, improving automated diagnostic accuracy from 20% to 80%. Combine model evaluation, data quality, ETL, and cross-functional delivery to translate technical requirements into production-oriented AI capabilities. Research experience spans wireless-network intelligence, simulation, transfer learning, and experimental platforms, alongside teaching and mentorship.

Technical Expertise, Leadership, and Professional Strengths

Generative & Agentic AI: LLM applications, retrieval-augmented generation (RAG), prompt engineering, document ingestion, chunking, embeddings, semantic retrieval, structured information extraction, tool-using agents, agent orchestration, and grounded response generation

Microsoft AI Ecosystem: Microsoft Foundry, Azure OpenAI, Azure AI Services, Azure AI Search, Azure Machine Learning, Foundry models and agents, Azure Content Understanding, Semantic Kernel, and responsible AI concepts

LLM Frameworks & Models: Hugging Face Transformers, LangChain, LangGraph, OpenAI APIs, transformer architectures, fine-tuning concepts, NLP foundation models, prompt templates, structured outputs, and LLM evaluation

Data & AI Platforms: Azure Databricks, Apache Spark, PySpark, Azure Data Factory, Azure Synapse Analytics, Azure Functions, Azure Blob Storage, Azure Data Lake, Google BigQuery, and cloud analytics workflows

Data Engineering: ETL/ELT pipelines, AI-ready datasets, batch processing, ingestion, cleansing, schema transformation, validation, feature engineering, data-quality assessment, and structured and unstructured data workflows

Machine Learning: PyTorch, TensorFlow, scikit-learn, XGBoost, transfer learning, domain adaptation, adversarial learning, anomaly detection, classification, regression, time-series forecasting, cross-validation, and model evaluation

MLOps & Software Engineering: Docker, Kubernetes, OpenShift, Git, GitHub Actions, CI/CD, REST APIs, Flask, testing, Linux, shell scripting, model deployment, reproducible workflows, and cloud-native application development

Programming & Analytics: Python, SQL, C/C++, and R; Pandas, NumPy, SciPy, Statsmodels, Matplotlib, Plotly, Streamlit, Jupyter, and exploratory data analysis

Wireless Systems: LTE/5G-NR, NR-NTN, cell selection, user timing in 5G standalone networks, RAN optimization, mobility, propagation modeling, and service assurance

Wireless AI & Optimization: Transfer learning, domain adaptation, adversarial learning, autoencoder-based CSI compression, GAN-based data generation, reinforcement learning, self-healing, and outage management

Simulation & Experimentation: Python simulation platforms, 3GPP-compliant network simulation, AI-enabled 5G testbeds, diagnostic logs, measurement analysis, robustness testing, and systematic experimentation

Leadership and Professional Strengths

- **Ownership and accountability:** Lead AI/ML initiatives through requirements analysis, data preparation, model development, validation, stakeholder review, and deployment planning.
- **Technical leadership:** Define technical approaches, evaluate alternatives, coordinate experimentation, and communicate findings to guide projects from ambiguous requirements to measurable outcomes.
- **Cross-functional collaboration:** Work with software engineers, network specialists, researchers, geologists, and public-safety stakeholders to translate domain requirements into practical data and AI solutions.
- **Continuous learning:** Apply ongoing study in generative AI, agentic AI, RAG, LLM evaluation, and cloud-native platforms through hands-on projects and professional coursework.
- **Reliability and commitment:** Maintain reproducible workflows, careful validation, and attention to detail throughout experimentation and delivery.
- **Adaptability and problem solving:** Combine critical thinking, systematic experimentation, and data-driven analysis to address unfamiliar technologies and complex operational problems.

Professional Experience

Machine Learning Engineer

AT&T, contractor via Innovatix Inc.

Feb. 2025-Present

Bedminster, NJ

- Own end-to-end development, testing, and operational integration of an AI-driven diagnostic system for critical E911 service-assurance workflows, improving automated root-cause classification accuracy from 20% to 80%.
- Designed domain-adaptive neural models and reusable data-processing workflows that generalize across heterogeneous operational domains and reduce performance degradation caused by distribution shifts.
- Developed multimodal AI prototypes combining work-order text, location information, network context, and operational signals to support GenAI-assisted diagnostics for VoNR and 5G service assurance.
- Created model-validation, error-analysis, and failure-mode assessment procedures to compare performance across domains and support reliable deployment in high-impact operational environments.
- Partnered with network engineers, software teams, and public-safety stakeholders to translate operational requirements into validated AI workflows and production-oriented data assets.

Data Scientist and AI/ML Engineer <i>Impac Exploration Services</i> - Built an LLM-enabled analytical pipeline for extracting and interpreting information from geochemical well data. - Developed scalable ingestion, cleansing, transformation, and validation workflows using Python, Azure Databricks, Azure Functions, and Synapse Analytics. - Automated ETL processing for large exploration datasets, reducing data-preparation time by 40% and improving reproducibility. - Implemented cross-validation, data-quality, and consistency checks that improved reliability of analytical outputs by 20%. - Collaborated with geologists to convert domain requirements and technical information into AI-ready datasets and actionable analytical outputs.	Sep. 2024-Dec. 2024 <i>Weatherford, OK</i>
Advanced-Degrees Intern <i>AT&T Labs</i> - Analyzed 500K+ LTE/5G-NR diagnostic logs and applied ML methods to detect anomalies, reducing diagnostic time by 20%. - Investigated 5G NR-NTN cell-selection behavior and improved decision algorithms by 15% through data-driven analysis and validation. - Conducted structured analysis of large operational datasets and communicated technical findings to engineering teams supporting next-generation wireless systems.	Jun. 2024-Aug. 2024 <i>Austin, TX</i>
Advanced Wireless Technology Development Intern <i>Futurewei Technologies</i> - Improved autoencoder-based channel-state-information compression by 25% using transfer learning and systematic experimentation on large wireless datasets. - Conducted 50+ experiments to validate model performance, robustness, and generalization under changing data conditions. - Collaborated with research and engineering stakeholders to translate modeling results into recommendations for next-generation communication systems.	Sep. 2022-Jan. 2023 <i>Schaumburg, IL</i>
Advanced-Degrees Intern <i>AT&T Labs</i> - Designed ML and statistical methods for analyzing users' timing in 5G standalone networks, improving analytical precision by 20%. - Analyzed large-scale user trace data with SQL and visualization tools and communicated findings to senior technical leadership. - Converted raw operational traces into interpretable performance indicators to support diagnosis and network optimization.	Jun. 2022-Aug. 2022 <i>Bedminster, NJ</i>
Graduate Research Assistant <i>AI4Networks, University of Oklahoma</i> - Designed reusable Python simulation and data-generation platforms for large-scale experimentation, including a 3GPP-compliant network simulator and an AI-enabled 5G testbed. - Built transfer-learning, adversarial-learning, GAN, autoencoder, and reinforcement-learning pipelines for sparse-data and cross-domain modeling. - Developed robustness evaluation frameworks for models operating under distribution shifts, missing data, and limited labeled samples; led 100+ experiments from problem formulation through validation. - Developed scalable self-healing and outage-management frameworks that improved modeled network resilience by 30%. - Published peer-reviewed research, presented technical results internationally, and mentored junior researchers in ML implementation, experimental design, and technical writing.	Sep. 2019-Sep. 2024 <i>Tulsa, OK</i>
Lecturer <i>Engineering and Technology Programs</i> Taught and mentored undergraduate engineering students in communications, programming, and foundational engineering subjects.	
Education	
PhD, Electrical and Computer Engineering University of Oklahoma, Norman, Oklahoma, USA	09/2019-07/2025
MSc, Telecommunication Engineering University of Engineering and Technology, Taxila, Pakistan	08/2014-08/2016
BSc, Telecommunication Engineering University of Engineering and Technology, Taxila, Pakistan	11/2010-06/2014
Selected Projects	
System Design Case-Study Portfolio <i>Guided Project</i> Documented architecture studies that translate product requirements into scalable services, data models, APIs, caching strategies, and reliability tradeoffs. <i>Tools & methods:</i> System Design, APIs, Distributed Systems, Scalability	2026 / In progress
Agentic AI and RAG Applications <i>Portfolio Project</i> Practical experiments with retrieval, tool-using agents, orchestration, memory, evaluation, and multi-agent workflows. <i>Tools & methods:</i> LangGraph, CrewAI, AutoGen, RAG, Vector Databases	2026 / In progress

Financial ML and Forecasting

2024-2025

Portfolio Project

- Credit-risk and loan-default modeling alongside time-series forecasting workflows for financial data.

Tools & methods: Python, Machine Learning, Time Series, Model Evaluation

Cloud Computer Vision

2025

Guided Project

- Real-time object detection, vehicle counting, and image-classification workflows using cloud-native AI services.

Tools & methods: Computer Vision, Streaming, Google Cloud AutoML, Python

AI-Ready Lakehouse Pipeline

2024

Portfolio Project

- An Azure Databricks data pipeline designed for analytics and downstream machine-learning workloads.

Tools & methods: Azure, Databricks, Spark, Data Engineering

Automated Meeting Minutes Generator with LLMs

2025

Guided Project

- A workflow that transcribes meetings, produces structured summaries, and extracts decisions and action items for downstream collaboration.

Tools & methods: Python, Speech-to-Text, LLM Summarization, Workflow Automation

LLM-Based Extraction from Unstructured Documents

2025

Guided Project

- Document intelligence pipeline for extracting entities, fields, and tables from reports, invoices, and contracts.

Tools & methods: LLMs, OCR, Named Entity Recognition, Data Integration

SyntheticNET: 3GPP-Compliant Network Simulation

2020-2024

Research Project

- Python-based simulation environment for realistic mobility, propagation, and data-driven experimentation in cellular networks.

Tools & methods: Python, 3GPP, Simulation, Wireless Networks

TurboRAN Experimental Platform

2020-2024

Research Infrastructure

- Multi-band, multi-tier experimental platform supporting evaluation of emerging AI-enabled radio-access-network solutions.

Tools & methods: 5G Testbeds, RAN, Experiment Design, Network Measurement

Wearable Intelligence for Health Monitoring

2024-2026

Research Project

- Deep-learning methods for continuous recognition of medication and dietary activities from wrist-worn inertial sensors.

Tools & methods: Wearable AI, Time Series, Deep Learning, Smartwatch Sensing

Certifications and Continuing Education

1. AWS Generative AI and AI Agents with Amazon Bedrock — [AWS/Coursera](#) | Completed (3/3) courses | 08/2026 | [Certificate](#)

Focus: Three-course professional certificate focused on developing, customizing, optimizing, and automating generative AI applications and intelligent agents with Amazon Bedrock.

Tools & Skills: Amazon Bedrock, Generative AI, AI Agents, Foundation Models, Knowledge Bases, Retrieval-Augmented Generation, LangChain, Amazon Q Developer, Model Customization, Prompt Engineering, Natural Language Processing, Text Generation, Summarization, Generative AI Application Development

2. IBM Generative AI Engineering Professional Certificate — IBM / Coursera | Completed (13/16) courses

Focus: Sixteen-course professional certificate developing job-ready skills across AI foundations, Python application development, machine learning, deep learning, NLP, transformer architectures, LLM fine-tuning, RAG, LangChain, and AI agents.

Tools & Skills: Generative AI Engineering, Large Language Models, Transformer Architectures, Prompt Engineering, RAG, LangChain, AI Agents, Hugging Face, PyTorch, Keras, Scikit-learn, Python, Flask, Gradio, PEFT, LoRA, QLoRA, RLHF, DPO, PPO, NLP, Machine Learning, Deep Learning, Vector Databases

3. IBM RAG and Agentic AI Professional Certificate — [IBM / Coursera](#) | Completed (8/8) courses | February 2026 | [Certificate](#)

Original eight-course credential.

Focus: Completed eight-course professional certificate covering retrieval-augmented generation, vector databases, multimodal applications, and agentic AI workflows.

Tools & Skills: Retrieval-Augmented Generation, Agentic AI, LangChain, LangGraph, Vector Databases, ChromaDB, FAISS, LlamaIndex, Multimodal AI, Tool Calling, Prompt Engineering, CrewAI, AutoGen / AG2, BeeAI, Gradio, Python, Flask

4. Generative AI Fundamentals Specialization — [IBM / Coursera](#) | Completed (5/5) courses | July 2026 | [Certificate](#)

Focus: Five-course foundation in generative AI, prompt engineering, LLMs, and practical enterprise applications.

Tools & Skills: Generative AI, Prompt Engineering, Foundation Models, Responsible AI, IBM watsonx, Hugging Face, Business Applications

5. Building AI Agents and Agentic Workflows Specialization — [IBM / Coursera](#) | Completed (3/3) courses | February 2026 | [Certificate](#)

Focus: Design agentic workflows with memory, conditional logic, tool use, and multi-agent orchestration using LangGraph, CrewAI, AutoGen, and BeeAI.

Tools & Skills: Agentic AI, LangGraph, LangChain, CrewAI, AutoGen / AG2, BeeAI, Agent Memory, Tool Calling, Multi-Agent Orchestration

- 6. RAG for Generative AI Applications Specialization** — [IBM / Coursera](#) | Completed (4/4) courses | March 2026 | [Certificate](#)
Focus: Build retrieval-augmented applications using LangChain, LlamaIndex, FAISS, and ChromaDB to connect language models with relevant external information.
Tools & Skills: Retrieval-Augmented Generation, LangChain, LlamaIndex, Vector Databases, FAISS, ChromaDB, Semantic Retrieval, LLM Applications
- 7. System Design Masterclass** — Packt / Coursera | Target: 2026
Focus: Architecture case studies covering scalable feeds, distributed data flows, caching, consistency, availability, and engineering tradeoffs.
Tools & Skills: System Design, Distributed Systems, Scalability, Caching
- 8. IBM DevOps and Software Engineering Professional Certificate** — [IBM / Coursera](#) | Completed (10/15) courses
Focus: A 15-course professional certificate covering software engineering, DevOps culture, Agile and Scrum, Python application development, Linux, Git and GitHub, cloud-native architecture, containers, microservices, serverless computing, automated testing, CI/CD, application security, monitoring, and observability.
Tools & Skills: DevOps, Software Engineering, Cloud-Native Development, CI/CD, Docker, Kubernetes, OpenShift, Microservices, Serverless Computing, Python, Linux, Bash, Git, GitHub, Agile Development, Scrum, Test-Driven Development, Behavior-Driven Development, Application Security, Monitoring and Observability, Prometheus, Grafana, Infrastructure as Code
- 9. Machine Learning** — Stanford University / Coursera | Completed
Focus: Core supervised and unsupervised learning, model evaluation, optimization, and practical ML system development.
Tools & Skills: Supervised Learning, Unsupervised Learning, Model Evaluation, Optimization
- 10. Machine Learning and Reinforcement Learning in Finance Specialization** — [New York University / Coursera](#) | Completed (0/4) courses
Focus: An applied machine-learning specialization focused on financial prediction, portfolio management, trading, risk modeling, market dynamics, and reinforcement-learning methods for finance.
Tools & Skills: Machine Learning for Finance, Reinforcement Learning, Financial Modeling, Portfolio Management, Financial Trading, Risk Modeling, Predictive Modeling, Supervised Learning, Unsupervised Learning, Statistical Machine Learning, Market Dynamics, Derivatives, TensorFlow, Scikit-learn, Python
- 11. Deep Learning for Healthcare Specialization** — [University of Illinois Urbana-Champaign / Coursera](#) | Completed (0/3) courses
Focus: An advanced specialization covering health-data processing, neural-network methods, and the development of deep-learning solutions for real-world medical and healthcare applications.
Tools & Skills: Deep Learning, Healthcare AI, Health Informatics, Health Data Processing, Artificial Neural Networks, Convolutional Neural Networks, Recurrent Neural Networks, Autoencoders, Generative Models, Medical Image Analysis, Supervised Learning, Unsupervised Learning, Predictive Modeling, Model Evaluation, PyTorch, Jupyter Notebooks
- 12. Google Cloud Data Analytics Professional Certificate** — Google Cloud / Coursera | Completed
Focus: Cloud data transformation, analysis, visualization, and decision support using Google Cloud tools.
Tools & Skills: SQL, BigQuery, Data Analytics, Visualization
- 13. Google Cloud Cybersecurity Professional Certificate** — Google Cloud / Coursera | Completed (5/5) courses | June 2026 | [Certificate](#)
Focus: Cloud-security training covering risk management, threat prevention, detection, incident response, and recovery in Google Cloud environments.
Tools & Skills: Google Cloud Security, Identity and Access Management, Threat Detection, Incident Response, Risk Management, Security Command Center, Cloud Logging, Vulnerability Management, Security Governance, Data Protection
- 14. AI for Telecommunications Specialization** — [AI CERTs / Coursera](#) | Completed (3/3) courses | June 2026 | [Certificate](#)
Focus: Three-course specialization connecting AI foundations with telecommunications network optimization, security, and customer experience.
Tools & Skills: AI for Telecommunications, Network Optimization, Cybersecurity, Customer Experience, Internet of Things, Python, TensorFlow
- 15. 4G Network Fundamentals** — [Institut Mines-Télécom / Coursera](#) | Completed (1/1) courses | August 2026 | [Certificate](#)
Focus: Foundational study of fourth-generation cellular networks.
Tools & Skills: 4G Networks, Cellular Networks, Telecommunications
- 16. Business Considerations for 5G with Edge, IoT, and AI** — [EDUCBA / Coursera](#) | Completed (1/1) courses | February 2026 | [Certificate](#)
Focus: Explores business considerations for combining 5G connectivity with edge computing, IoT, and AI.
Tools & Skills: 5G, Edge Computing, Internet of Things, Artificial Intelligence, Business Analysis

Selected Publications

Real-World Medication Adherence Monitoring Using Hybrid Neural Networks and Smartwatches <i>HealthINF Accepted</i>	2026
On Multi-Parameter Optimization and Proactive Reliability in Cellular Networks <i>Journal article</i>	2025
An AI-Driven Framework for Enhancing Resilience in Propagation Models to Enable Digital Twin <i>IEEE PIMRC</i>	2024

Data-Driven Intelligent Outage Management for High Shadowing Environments in 5G&B Networks <i>IEEE SmartNets</i>	2024
Toward a Hybrid RF/Optical Lunar Communication System (LunarComm) <i>IEEE Network</i>	2022
Towards Positioning Error Impact Characterization and Minimization in User-Centric RAN <i>IEEE WCNC</i>	2022
Unification of RF Energy Harvesting Schemes under Mixed Rayleigh-Rician Fading Channels <i>AEÜ - International Journal of Electronics and Communications</i>	2020